

Before class: from atoms to ions

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

About 15 minutes. Read the three short cards, then try the readiness check. Wrong answers are fine: they show what to listen for in the lecture.

Read the cards first, then answer the questions without notes. A periodic table and the answer key are at the back.

1 · Read first (5 min)

Why do atoms bond?

The noble gases (He, Ne, Ar ...) hardly ever form compounds. Apart from helium, they all have **eight valence electrons**. Other atoms become more stable by reaching eight valence electrons too: the **octet rule**. Very small atoms (H, Li) are stable with two (the **duet rule**).

An atom can reach an octet by **transferring** electrons (this lecture) or by **sharing** them (the next lecture, IC2).

What is an ion?

- An atom is neutral: protons (+) = electrons (–).
- If an atom **loses** electrons it has more protons than electrons: a positive ion, a **cation**. Metals usually do this.
- If an atom **gains** electrons it becomes a negative ion, an **anion**. Nonmetals usually do this.
- Opposite charges attract. That attraction is an **ionic bond**, and the substance is an **ionic compound**, such as sodium chloride (NaCl).

Why this matters in pharmacy

Normal saline (0.9% NaCl) is given intravenously every day in hospitals. Iron tablets (ferrous sulfate), antacids (calcium carbonate, magnesium hydroxide) and Epsom salt (magnesium sulfate) are all ionic compounds. In this lecture you will learn to write their formulas, name them and calculate their masses.

2 · Readiness check

Ten quick questions on what you need for today.

Q1 **Recap**

How many valence electrons does a sodium atom (Group 1A) have?

- A. 1
- B. 2
- C. 7
- D. 8

Q2 **Recap**

A chlorine atom (Group 7A) has ____ valence electrons.

Q3 **IC1-LO1**

Why are the noble gases so unreactive?

- A. They are very heavy
- B. Their valence shell is full (an octet, or a duet for He)
- C. They have no electrons
- D. They are always ions

Q4 **Recap**

Classify each element.

Options: metal · nonmetal

magnesium (Mg)

sulfur (S)

potassium (K)

oxygen (O)

Q5 IC1-LO2

An ion has 11 protons and 10 electrons. What is its charge?

- A. 1–
- B. 0
- C. 1+
- D. 2+

Q6 IC1-LO2

Metals usually lose electrons and become positive ions (cations).

True / False

Q7 IC1-LO2

Is a sodium atom more likely to lose 1 electron or gain 7 to reach an octet?

- A. Lose 1
- B. Gain 7
- C. Neither
- D. Both at once

Q8 IC1-LO2

Oxygen (Group 6A) gains ____ electrons to complete its octet.

Q9 IC1-LO1

An ionic compound can be made of two positive ions.

True / False

Q10 IC1-LO3

Which pair of elements is most likely to form an **ionic** compound?

- A. C and O
- B. N and H
- C. Ca and Cl
- D. S and O

3 • Think ahead

No wrong answers here. Make a guess, then check it after the lecture (Parts 1-2).

Q11 IC1-LO1

Sodium is a reactive metal and chlorine is a poisonous gas. Why is sodium chloride safe enough to eat and to give in an IV drip?

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Periodic table · each box: atomic number, symbol, atomic mass, **electronegativity** · colours: metal, metalloid, nonmetal, noble gas

1A 1	2A 2	3	4	5	6	7	8	9	10	11	12	3A 13	4A 14	5A 15	6A 16	7A 17	8A 18
1 H 1.01 2.1																	2 He 4.00 —
3 Li 6.94 1.0	4 Be 9.01 1.5											5 B 10.81 2.0	6 C 12.01 2.5	7 N 14.01 3.0	8 O 16.00 3.5	9 F 19.00 4.0	10 Ne 20.18 —
11 Na 22.99 0.9	12 Mg 24.31 1.2											13 Al 26.98 1.5	14 Si 28.09 1.8	15 P 30.97 2.1	16 S 32.07 2.5	17 Cl 35.45 3.0	18 Ar 39.95 —
19 K 39.10 0.8	20 Ca 40.08 1.0	21 Sc 44.96 1.3	22 Ti 47.87 1.5	23 V 50.94 1.6	24 Cr 52.00 1.6	25 Mn 54.94 1.5	26 Fe 55.84 1.8	27 Co 58.93 1.9	28 Ni 58.69 1.9	29 Cu 63.55 1.9	30 Zn 65.38 1.6	31 Ga 69.72 1.6	32 Ge 72.63 1.8	33 As 74.92 2.0	34 Se 78.97 2.4	35 Br 79.90 2.8	36 Kr 83.80 —
37 Rb 85.47 0.8	38 Sr 87.62 1.0	39 Y 88.91 1.2	40 Zr 91.22 1.4	41 Nb 92.91 1.6	42 Mo 95.95 1.8	43 Tc [98] 1.9	44 Ru 101.07 2.2	45 Rh 102.91 2.2	46 Pd 106.42 2.2	47 Ag 107.87 1.9	48 Cd 112.41 1.7	49 In 114.82 1.7	50 Sn 118.71 1.8	51 Sb 121.76 1.9	52 Te 127.60 2.1	53 I 126.90 2.5	54 Xe 131.29 —
55 Cs 132.91 0.7	56 Ba 137.33 0.9	57–71	72 Hf 178.49 1.3	73 Ta 180.95 1.5	74 W 183.84 1.7	75 Re 186.21 1.9	76 Os 190.23 2.2	77 Ir 192.22 2.2	78 Pt 195.08 2.2	79 Au 196.97 2.4	80 Hg 200.59 1.9	81 Tl 204.38 1.8	82 Pb 207.2 1.9	83 Bi 208.98 1.9	84 Po [209] 2.0	85 At [210] 2.2	86 Rn [222] —
87 Fr [223] 0.7	88 Ra [226] 0.9	89–103	104 Rf [267] —	105 Db [268] —	106 Sg [269] —	107 Bh [270] —	108 Hs [277] —	109 Mt [278] —	110 Ds [281] —	111 Rg [282] —	112 Cn [285] —	113 Nh [286] —	114 Fl [289] —	115 Mc [290] —	116 Lv [293] —	117 Ts [294] —	118 Og [294] —
Lanthanides		57 La 138.91 —	58 Ce 140.12 —	59 Pr 140.91 —	60 Nd 144.24 —	61 Pm [145] —	62 Sm 150.36 —	63 Eu 151.96 —	64 Gd 157.25 —	65 Tb 158.93 —	66 Dy 162.50 —	67 Ho 164.93 —	68 Er 167.26 —	69 Tm 168.93 —	70 Yb 173.05 —	71 Lu 174.97 —	
Actinides		89 Ac [227] 1.1	90 Th 232.04 1.3	91 Pa 231.04 1.4	92 U 238.03 1.4	93 Np [237] —	94 Pu [244] —	95 Am [243] —	96 Cm [247] —	97 Bk [247] —	98 Cf [251] —	99 Es [252] —	100 Fm [257] —	101 Md [258] —	102 No [259] —	103 Lr [262] —	

Atomic masses to 2 d.p. ([n] = mass number of the most stable isotope). Electronegativity: Pauling values as in Ball, *The Basics of GOB Chemistry*, Figure 4.4.3 (La–Lu 1.0–1.2; noble gases have no value).

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About 15 minutes. Read the three short cards, then try the readiness check. Wrong answers are fine: they show what to listen for in the lecture.

Answer key

2 · Readiness check

Q1 **Recap**

How many valence electrons does a sodium atom (Group 1A) have?

A. 1

B. 2

C. 7

D. 8

Group 1A → 1 valence electron.

Q2 **Recap**

A chlorine atom (Group 7A) has ____ valence electrons.

7

Group 7A → 7.

Q3 **IC1-LO1**

Why are the noble gases so unreactive?

A. They are very heavy

B. Their valence shell is full (an octet, or a duet for He)

C. They have no electrons

D. They are always ions

A full valence shell is especially stable, so they have little reason to gain, lose or share electrons.

Q4 Recap

Classify each element.

magnesium (Mg) → metal; sulfur (S) → nonmetal; potassium (K) → metal; oxygen (O) → nonmetal

Q5 IC1-LO2

An ion has 11 protons and 10 electrons. What is its charge?

- A. 1−
- B. 0
- C. 1+**
- D. 2+

11 positive − 10 negative = 1+.

Q6 IC1-LO2

Metals usually lose electrons and become positive ions (cations).

True / False → **True**

Metals have few valence electrons, which they lose easily.

Q7 IC1-LO2

Is a sodium atom more likely to lose 1 electron or gain 7 to reach an octet?

- A. Lose 1**
- B. Gain 7
- C. Neither
- D. Both at once

Losing one electron is far easier than gaining seven. The result is Na^+ .

Q8 IC1-LO2

Oxygen (Group 6A) gains ____ electrons to complete its octet.

2

$6 + 2 = 8$, giving the oxide ion O^{2-} .

Q9 IC1-LO1

An ionic compound can be made of two positive ions.

True / False → **False**

Like charges repel. An ionic compound always contains both cations and anions.

Q10 IC1-LO3

Which pair of elements is most likely to form an **ionic** compound?

A. C and O

B. N and H

C. Ca and Cl

D. S and O

Calcium is a metal and chlorine a nonmetal: electrons are transferred. The others are nonmetal pairs (covalent, IC2).

3 • Think ahead

Q11 IC1-LO1

Sodium is a reactive metal and chlorine is a poisonous gas. Why is sodium chloride safe enough to eat and to give in an IV drip?

When sodium and chlorine react, each sodium atom **transfers** one electron to a chlorine atom. Both become ions with stable octets (Na^+ and Cl^-). The compound's properties are completely different from those of the elements: the reactive metal and the toxic gas are gone, and the ions are held together by ionic bonds.

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